

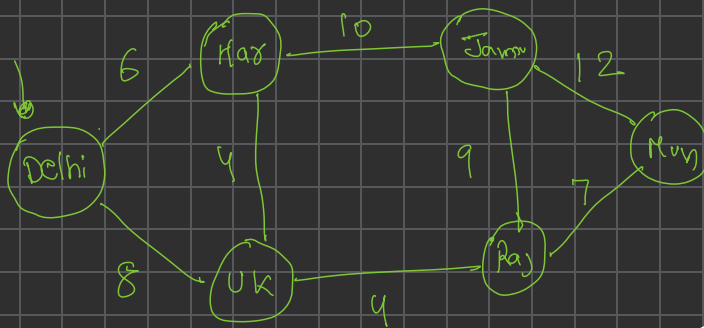
Algorithm

[Graph] $\Rightarrow$ Important

[Google maps]

node or vertex

edges

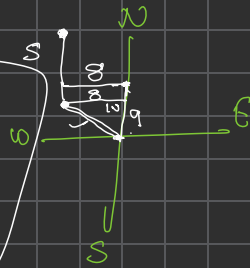


19
28
21

Del $\rightarrow$ UK $\rightarrow$ Raj $\rightarrow$ Mun

Del $\rightarrow$ UK = 8
UK $\rightarrow$ Raj = 4
Raj $\rightarrow$ M = 7

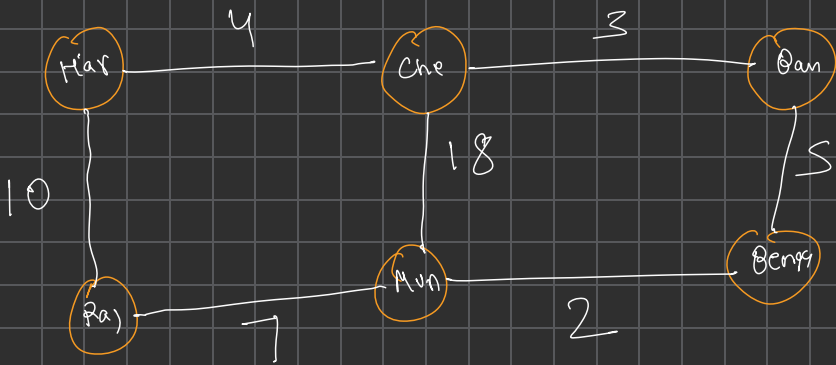
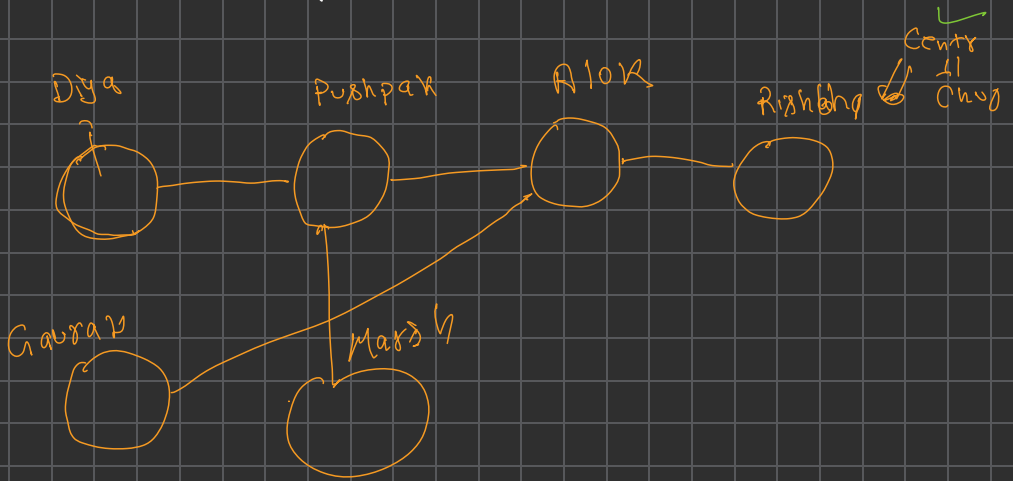
Del $\rightarrow$ Har = 6

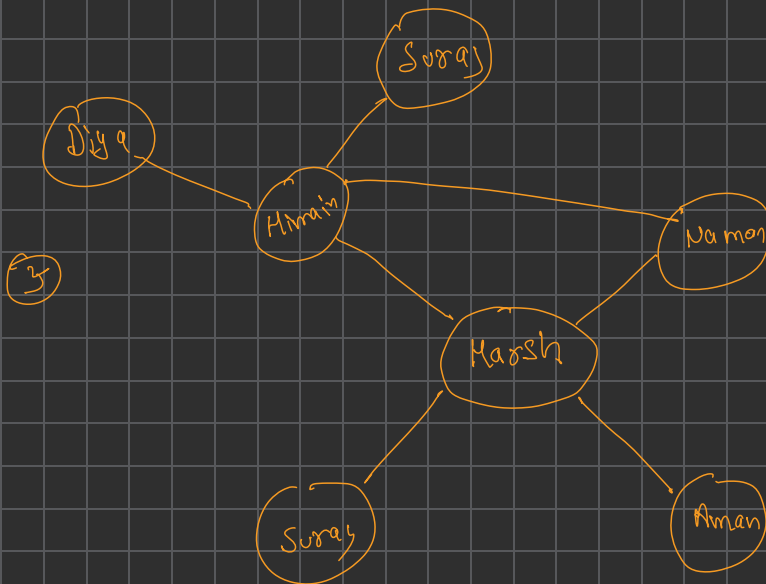
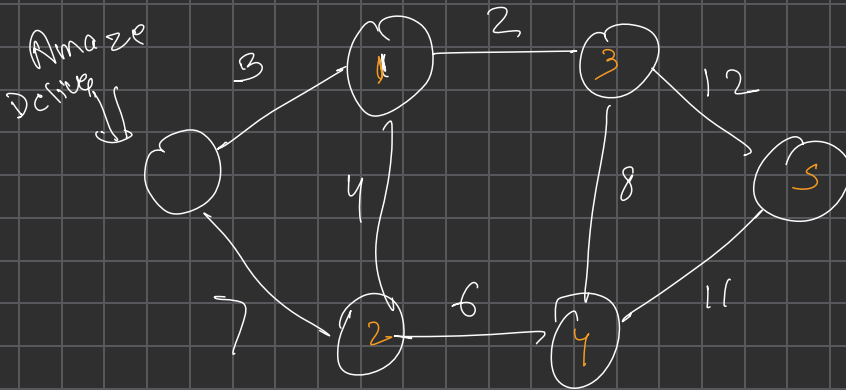


(1)

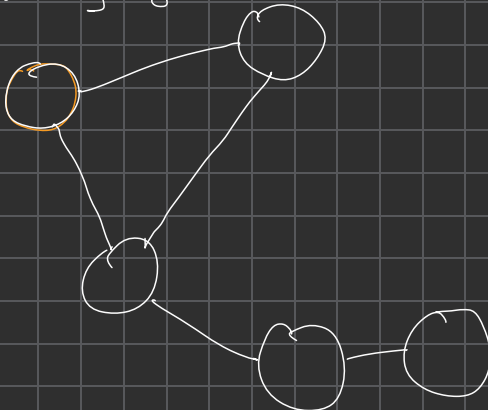
10 KM North, 90° left for 8 KM,
90 deg Right 5 KM, 180° U, 6 KM,
Start ki

→ Inspector: NEET - 2026 Criminal

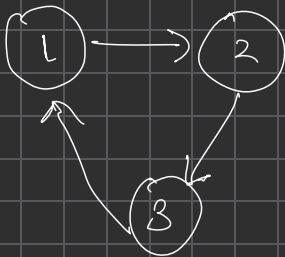




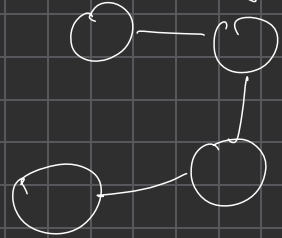
① Cyclic Graph
Undirected Cyclic Graph



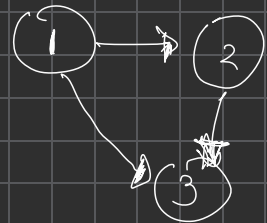
Directed cyclic graph

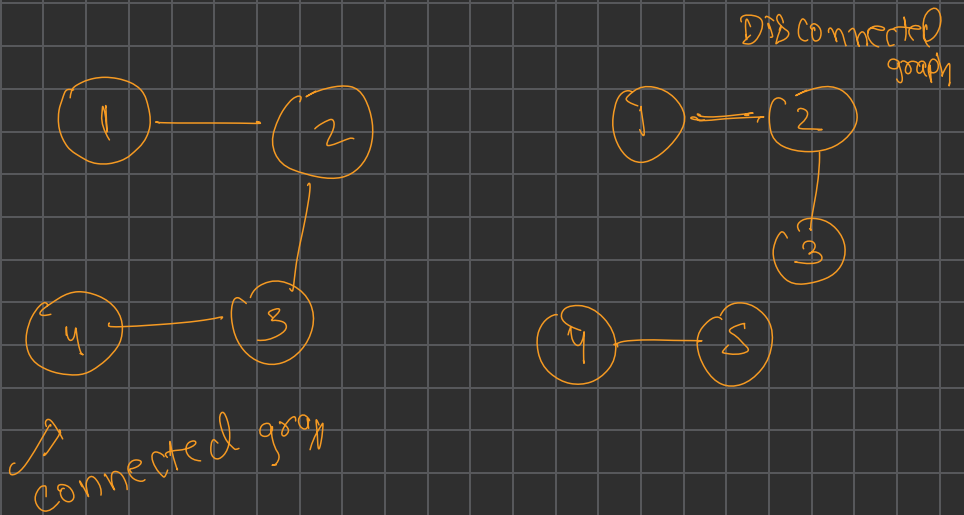


(vs) Acyclic Graph
Undirected acyclic graph

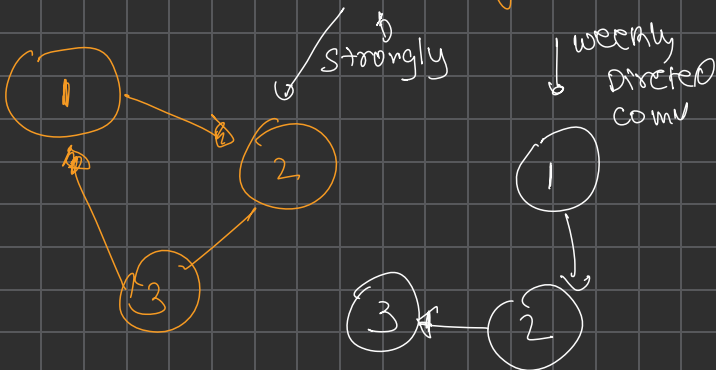


Directed acyclic graph





Directed connected graph



Directed disconnected gr



① Complete graph

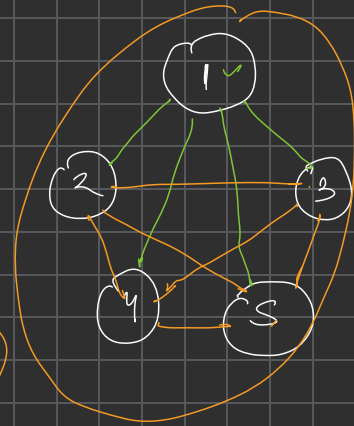
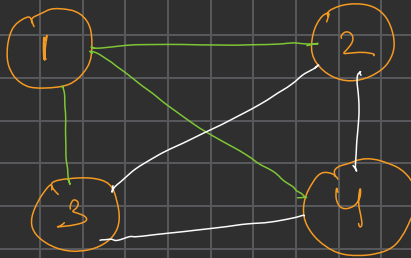
no of edges ✓

$$n = 4$$

$$ed = 6$$

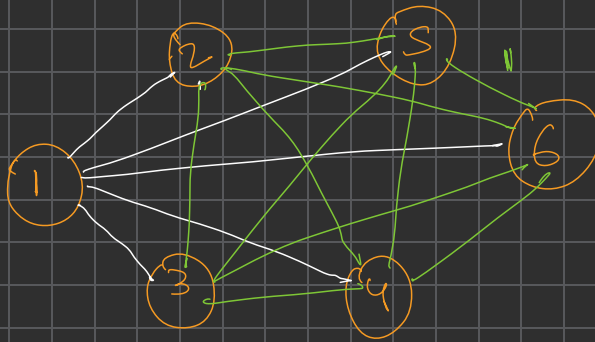
$$S = 5$$

$$edges = 10$$



$$n = 6$$

$$edges = 15$$



sum

$$1 + n - 1$$

$$1 \Rightarrow 5$$

$$2 \Rightarrow 4$$

$$3 \Rightarrow 3$$

$$4 \Rightarrow 2$$

$$5 \Rightarrow 1$$

$$6 \Rightarrow 0$$

$$n C_2 \Rightarrow$$

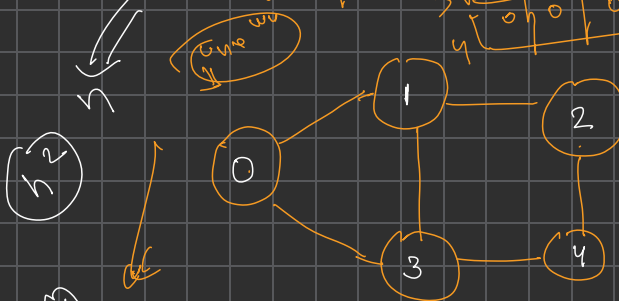
$$n \times (n+1) \Rightarrow \frac{(n-1)(n)}{2}$$

$$\frac{n \times (n-1)}{2}$$

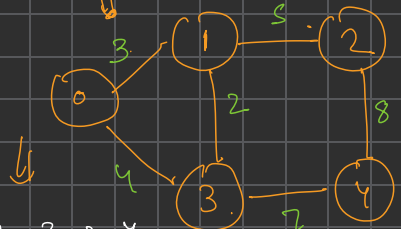
(Vertex)²
 vertex = 5
 edge = 8

	0	1	2	3	4
0	0	1	0	0	0
1	1	0	1	0	0
2	0	1	0	0	0
3	0	0	0	0	0
4	0	0	0	0	0

↓
 $V = 5$
 $E = 8$
 $V = 1$
 $V = 2$



System Data
 Array
 Link List



① Adj Matrix

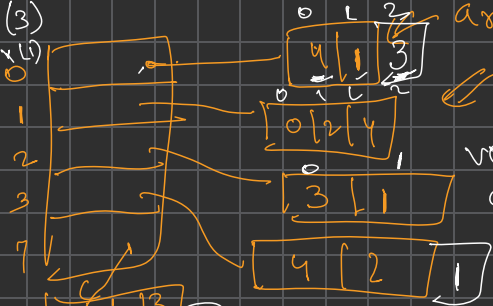
	0	1	2	3	4
0	0	1	0	1	0
1	1	0	1	1	0
2	0	1	0	0	1
3	1	1	0	0	1
4	0	0	1	1	0

1/8/10/10/10/10/10/10

	0	1	2	3	4
0	0	3	0	4	0
1	3	0	5	2	0
2	0	5	0	0	8
3	4	2	0	0	7
4	0	0	8	7	0

Space: (Vertex)²

adjList[0].push-back(3)
 adjList[3].push-back(1)

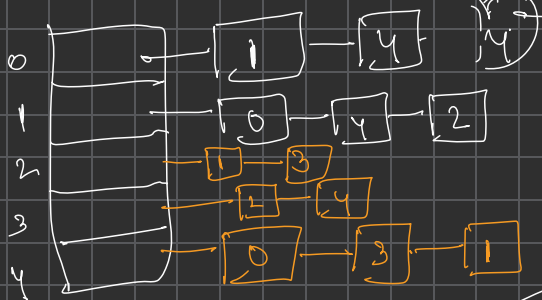


vector<vector<int>>
 adjList(5) ~

AdjList

vertex = V
 custom

Space save
 search
 $O(V)$



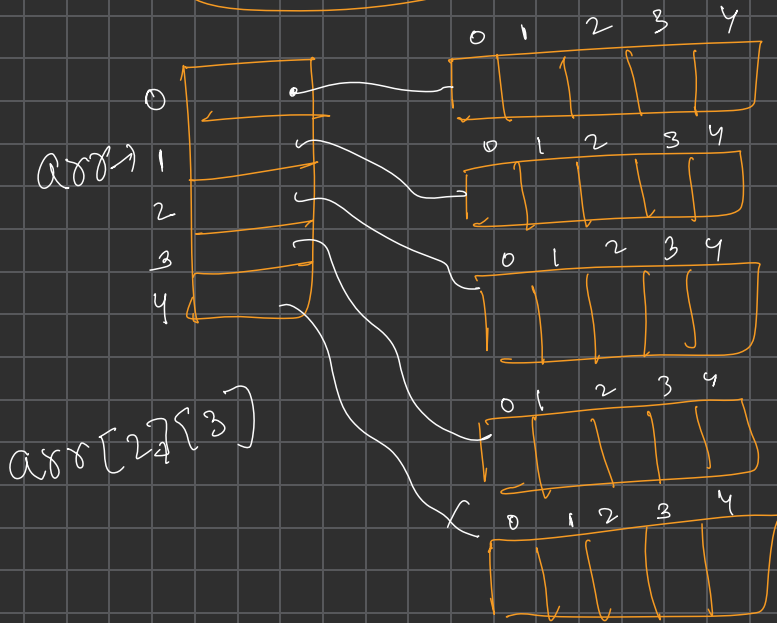
2d vector

2d array
 $O(V)$

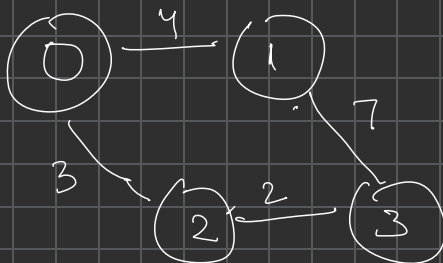
	0	1	
0	1	4	
1	0	4	2
2	1	3	
3	2	4	
4	0	3	1

2 d array

(5x5)

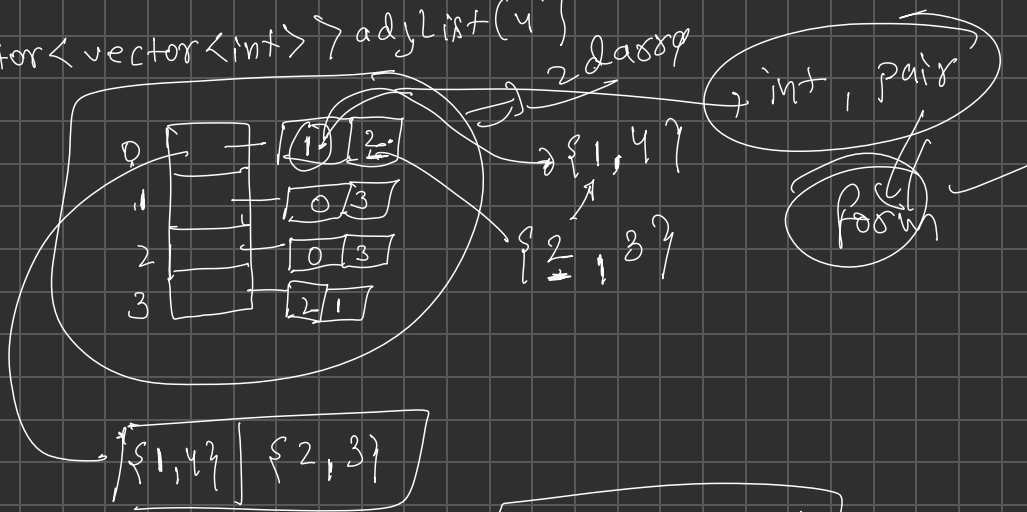


vertex = 4
edges = 4

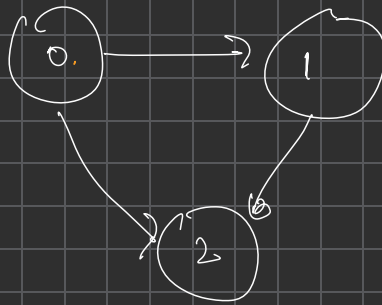


`adjList[0].push(1);`

`vector<vector<int>> adjList(4)`



Directed graph $\Rightarrow$ mat
unewig $\Leftarrow$ Adjacen



	0	1	2
0	0	1	1
1	0	0	1
2	1	0	0

Graph

Loop

Disconnected

Tree

No-Loop

Connected

